A Refined Hybrid Network for Object Detection in Aerial Images

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***Abstract*— Aerial object detection is a challenging task that needs to detect objects with large variations in scale and orientation. Previous dense object detectors rely on heuristic nonmaximum suppression (NMS) to filter out redundant detections. This may reduce the recall rate for objects with arbitrary orientations and large aspect ratios. Recently proposed sparse object detectors treat object detection as a set prediction task, effectively eliminating the need for hand-crafted components. However, applying this paradigm directly to aerial images achieves inferior performance. In this article, we develop an effective refined hybrid network (RHNet) for object detection in aerial images. Our method combines the advantages of both dense and sparse detectors, achieving outstanding performance for aerial objects with large variations. Specifically, considering the highly diverse orientations of objects, we first apply a dynamic query generation (DQG) module to produce high-quality oriented queries. These queries can effectively locate the foreground objects in an image, ensuring a high recall rate. Then, the object queries are sent to a query decoder for further refinement. This refinement stage adopts one-to-one matching to eliminate the negative impact caused by NMS. Moreover, an adaptive feature fusion (AFF) module is designed to learn stronger modeling capabilities for rotated objects at different scales. In addition, we propose a practical mixed query sampling (MQS) strategy that uses many-to-one assignment as an auxiliary scheme to help detector training. Extensive experiments conducted on several aerial datasets demonstrate the superior performance of the proposed method in comparison to other state-of-the-art approaches.**

***Index Terms*— Adaptive feature fusion (AFF), aerial images, dynamic query generation (DQG), hybrid network, mixed query sampling (MQS).**

I. Introduction

OBJECT detection in aerial images has become an increasingly important but challenging research topic in remote sensing. This task involves finding objects of interest in an image with accurate bounding boxes and identifying their categories, and it has broad real-world applications such as environment monitoring, traffic management, and urban planning. However, it is challenging as objects in aerial images often appear in arbitrary orientations and large-scale variations (see Fig. 1).

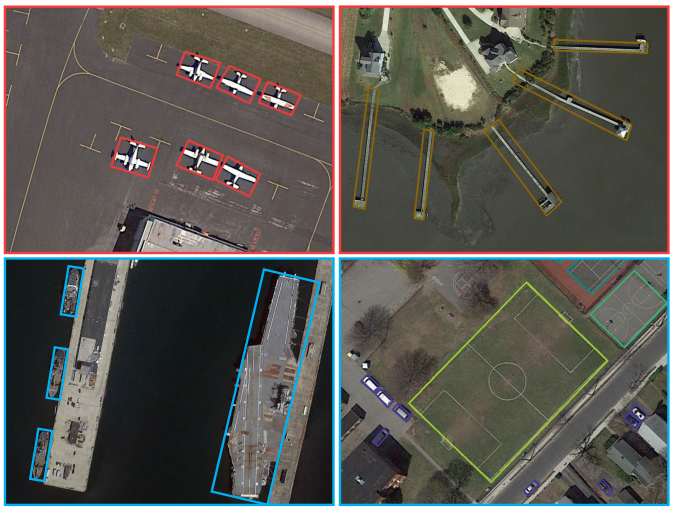


Fig. 1. Illustration of objects in aerial images. (Top) Two images show the objects with arbitrary orientations. (Bottom) Two images show objects with large-scale variations.

Owing to the advances in convolutional neural networks (CNNs), there are many well-established methods that facilitate this research and achieve remarkable performance [1], [2], [3], [4], [5]. These methods usually rely on dense prior information, i.e., anchor boxes or anchor points placed uniformly over the image to locate the objects with large variations, making heuristic nonmaximum suppression (NMS) postprocessing necessary to remove redundant boxes. This postprocessing inevitably hurts some correct predictions and limits the detection performance, especially for objects with arbitrary orientations and large aspect ratios. As shown in Table I, the mAP of FCOSR [5] is improved from 48.44% to 68.12% with NMS. Nevertheless, the recall rates of some categories drop at the same time, such as bridges (BD) and harbors (HA). It reveals that some accurately localized oriented boxes are wrongly suppressed.

Recently, sparse object detectors have drawn a lot of attention owing to their simple architecture and competitive performance [6], [7], [8], [9]. In general, these methods regard object detection as a set prediction task and replace dense priors with a sparse set of object queries learned by a one-to-one

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TABLE I
COMPARISON RESULTS WITH RELATED METHODS ON THE DOTA1.0 VALIDATION SET. THE TOP PART REPRESENTS AP OF EACH CATEGORY, AND THE BOTTOM PART REPRESENTS RECALL OF EACH CATEGORY

Methods	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP(%)
FCOSR (w/o NMS)	65.85	53.05	31.24	43.22	57.79	60.12	72.13	59.07	41.66	49.53	38.45	42.67	48.21	36.28	27.26	48.44
FCOSR (w/ NMS)	89.51	71.60	37.80	63.23	**74.03**	85.19	**88.71**	**90.62**	62.68	72.69	**60.85**	**63.17**	66.53	53.12	42.01	68.12
Sparse R-CNN	85.39	49.44	27.91	44.17	35.91	61.31	48.87	88.47	51.21	72.38	46.39	48.29	54.78	45.22	35.08	52.99
Selected queries	89.51	71.74	37.81	57.44	68.16	84.49	86.90	90.63	62.92	72.71	44.89	63.03	65.61	53.18	42.46	66.10
Hybrid network	**90.13**	**71.79**	**52.47**	**68.80**	72.02	**86.10**	88.53	90.55	**67.09**	**77.23**	51.23	60.70	**78.44**	**56.59**	**57.26**	**71.26**
FCOSR (w/o NMS)	97.12	91.34	81.40	90.09	**95.86**	**96.00**	**98.53**	**95.30**	87.97	84.87	91.63	**85.45**	**95.66**	**81.97**	87.70	-
FCOSR (w/ NMS)	95.26	85.47	62.80	84.43	91.98	95.61	96.55	93.85	80.45	80.91	82.87	81.09	80.85	74.04	79.51	-
Sparse R-CNN	92.08	77.51	65.09	85.50	54.75	76.64	61.47	94.86	72.58	**86.57**	**94.25**	81.71	72.98	71.04	75.00	-
Selected queries	95.64	87.99	64.84	77.36	84.94	93.97	90.96	94.91	86.09	81.20	63.35	81.82	80.61	75.14	83.61	-
Hybrid network	**97.53**	**93.30**	**84.59**	**94.81**	91.02	94.81	95.73	94.91	**92.48**	85.59	91.63	84.73	91.38	76.3	**91.80**	-

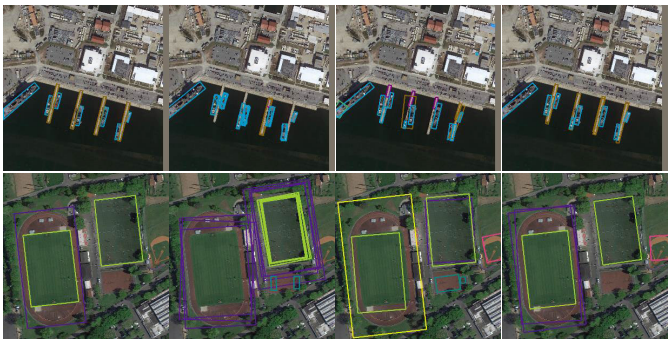


Fig. 2. Visualized detection results of different methods on the DOTA1.0 Validation Set. GT denotes GT objects.

matching loss. In this way, it eliminates the need for NMS postprocessing. However, due to the lack of prior knowledge and highly diverse directions of aerial objects, the learnable object queries struggle to cover all the oriented objects in an image, which also affects the performance of latter refinement stages. In Table I, Sparse R-CNN [7] only achieves 52.99% mAP, indicating that applying sparse detectors directly to aerial images achieves inferior performance.

In this article, we develop an effective refined hybrid network (RHNet) for object detection in aerial images. Our method combines the advantages of both dense and sparse object detectors, that is, maintaining high recall for arbitrary-oriented objects and eliminating NMS postprocessing, leading to superior detection performance. Specifically, we first apply a dynamic query generation (DQG) module that leverages dense priors to provide better initial oriented queries. This module generates dynamic object queries based on varying input images. Then, these queries are sent to subsequent query decoder for further refinement. And the refinement stage adopts one-to-one matching to eliminate the negative impact caused by NMS. From Table I, we see that the recall rate of selected queries is lower than FCOSR before NMS. After refinement, the recall is greatly improved and even exceeds the results of FCOSR before NMS, thus significantly improving the detection accuracy. Fig. 2 shows the comparison results of different methods. Note that we use a soft object-adaptation label assignment strategy in DQG to define the positive and negative samples, aiming at adapting to arbitrary shapes and directions of aerial objects. Moreover, considering the orientation and scale variations in objects, we propose an adaptive feature fusion (AFF) module to improve the robustness of feature representation, which can effectively use the feature maps of varying scales from local rotational regions of interest (RoIs). Besides, to better exploit more training samples for refinement stage, we introduce an auxiliary supervision that leverages the widely used many-to-one assignment scheme to assist the original one-to-one assignment. And a mixed query sampling (MQS) strategy is designed for our hybrid network, which selects two different groups of object queries for these two assignment schemes, respectively.

To validate the effectiveness of our method, we carry out experiments on several datasets: DOTA1.0 [10], HRSC2016 [11], and UCAS-AOD [12]. In addition, we perform in-depth ablation studies to explore the importance of different components in our design. The experimental results demonstrate that the proposed RHNet achieves competitive performance with the state-of-the-art methods.

In summary, the main contributions are given as follows.

1) We analyze the issues of the existing dense and sparse object detectors in aerial images and present a new RHNet to effectively combine the merits of these two types of detection paradigms, thus achieving superior detection performance.
2) We apply a DQG module to generate high-quality oriented queries covering the potential objects in an image. Subsequently, a query decoder is used to refine the queries with a one-to-one matching loss, which eliminates the negative impact caused by NMS.
3) Considering the orientation and scale variations in objects in aerial images, we design an AFF module to better exploit the multiscale rotational RoI features. Besides, we propose a practical MQS strategy to leverage the auxiliary many-to-one assignment for improving detector training.

II. RELATED WORK

A. Object Detection in Aerial Images

In recent years, CNN-based object detection has been well-studied and has made great progress. These methods can be roughly grouped into two types: two-stage detectors and one-stage detectors. In general, the two-stage detectors rely on region proposal networks (RPNs) to produce class-agnostic candidate boxes, which are then refined and classified into

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different categories in the second stage. There are some representative work such as Faster R-CNN [13] and Mask R-CNN [14]. In contrast, one-stage detectors such as RetinaNet [15] and FCOS [16] directly produce the predictions based on dense feature maps. The advancement of general object detection has brought significant improvements to research on aerial images. Considering that objects in aerial images often appear in arbitrary orientations and large-scale variations, directly applying general object detection methods usually results in inferior model performance. Therefore, lots of work have been done to study oriented object detection and has attained promising results. For instance, Ma et al. [17] replaced RPN in Faster R-CNN with a rotational RPN (RRPN) to obtain inclined proposals, which used plenty of rotated anchors with varying scales, aspect ratios, and angles to cover the oriented objects. To reduce the number of anchors and decrease the computation complexity, Ding et al. [1] proposed the RoI Transformer that learned the transformation for converting the horizontal RoIs into rotational proposals, and then performed classification and regression based on these proposals. An oriented RPN was further designed by Xie et al. [18] to produce the oriented proposals with much fewer parameters than RRPN and RoI Transformer. To better describe oriented objects, a novel oriented bounding box (OBB) representation was proposed by Xu et al. [4] that glided the vertex of horizontal bounding box on each side. Han et al. [19] incorporated a rotation-equivariant network into the detector, aiming to extract more accurate rotation-invariant features. Besides, there is some work focusing on one-stage or anchor-free frameworks. For instance, a refined one-stage detector was developed by Yang et al. [3] with a cascade structure to improve the performance. Han et al. [20] also used a feature alignment module to refine the convolutional features for corresponding rotational proposals in one-stage detectors. An anchor learning strategy for oriented object detection was designed by Ming et al. [21], which dynamically selected training samples for each object according to the newly defined matching degree. Considering the shape and direction of arbitrary-oriented objects, Huang et al. [22] proposed a more appropriate object-adaptation label assignment mechanism for anchor-free detectors that adopted oriented Gaussian heatmaps to define the positive samples. Yu et al. [23] also considered the weights for different training samples via a soft label assignment strategy and extracted the features for oriented objects by an oriented feature refinement module. Overall, these methods have led to a significant improvement in performance of object detection in aerial images. Nevertheless, they follow the paradigm of first generating dense predictions and then adopting NMS postprocessing to remove redundant boxes during inference. But this process cannot guarantee to perfectly filter out false-positive predictions while retaining all the true positives.

B. Sparse Object Detectors

Recently, Carion et al. [6] introduced a new paradigm of fully end-to-end object detection with transformers (DETRs), which generated a sparse set of predictions and adopted bipartite matching for label assignment, eliminating the need for postprocessing. Although DETR achieves comparable performance with Faster R-CNN, it suffers from poor detection performance for small objects and slow training convergence. Much follow-up work has been done to tackle these issues. For instance, Zhu et al. [8] designed a deformable attention module which attends to a small set of sampling points for each reference, significantly reducing the training time and achieving better performance than DETR. In addition, a purely sparse framework for object detection named Sparse R-CNN was proposed by Sun et al. [7], which used a dynamic instance interactive head to extract object features from local regions. A dynamic proposal generation module was introduced by Hong et al. [24], which used multiple experts to provide better initial proposals and improved the label matching process. Zhang et al. [25] explored the convergence of end-to-end object detection from the perspective of query design, and then proposed dense distinct queries to accelerate network training. However, these methods fail to consider the arbitrary shapes and orientations of aerial objects. Besides, some work tries to improve training efficiency and detection accuracy in other ways [26], [27], [28], [29]. For instance, a novel query denoising scheme was presented by Li et al. [26] that increased the stability of bipartite graph matching during training. Jia et al. [28] improved the training efficiency by introducing a hybrid matching strategy to combine the leverages of both one-to-one and many-to-one label assignments. Different from the learnable object queries in [28], the proposed MQS selects two different sparse sets of high-quality object queries from the dense predictions of DQG. Ma et al. [30] proposed a transformer-based method that applied sparse detector to oriented object detection for the first time. Dai et al. [31] further improved the performance of transformer networks for arbitrary-oriented objects. Nevertheless, it still takes a long time for the network to converge. The proposed method combines the advantages of both dense and sparse detectors, thus achieving promising detection performance and training efficiency on aerial object detection tasks.

III. Methodology

In this section, we illustrate the overall framework of the proposed RHNet for aerial images, as described in Fig. 3. Our RHNet mainly contains three parts: 1) the hybrid network; 2) the AFF module; and 3) the MQS strategy. The image features are extracted from the backbone with a feature pyramid network (FPN) [32] and then sent to subsequent stages for final prediction. The hybrid network contains a DQG module to produce dynamic object queries regarding the input image and a query decoder for further refinement. The AFF module is designed to effectively aggregate the information from image features based on the input query feature. Moreover, MQS strategy is adopted to select two different groups of queries for one-to-one and many-to-one label assignments, respectively. We elaborate on the components of our model in the following parts.

A. Hybrid Network

Previous dense object detectors rely on NMS postprocessing to remove redundant detections, which inevitably hurts some

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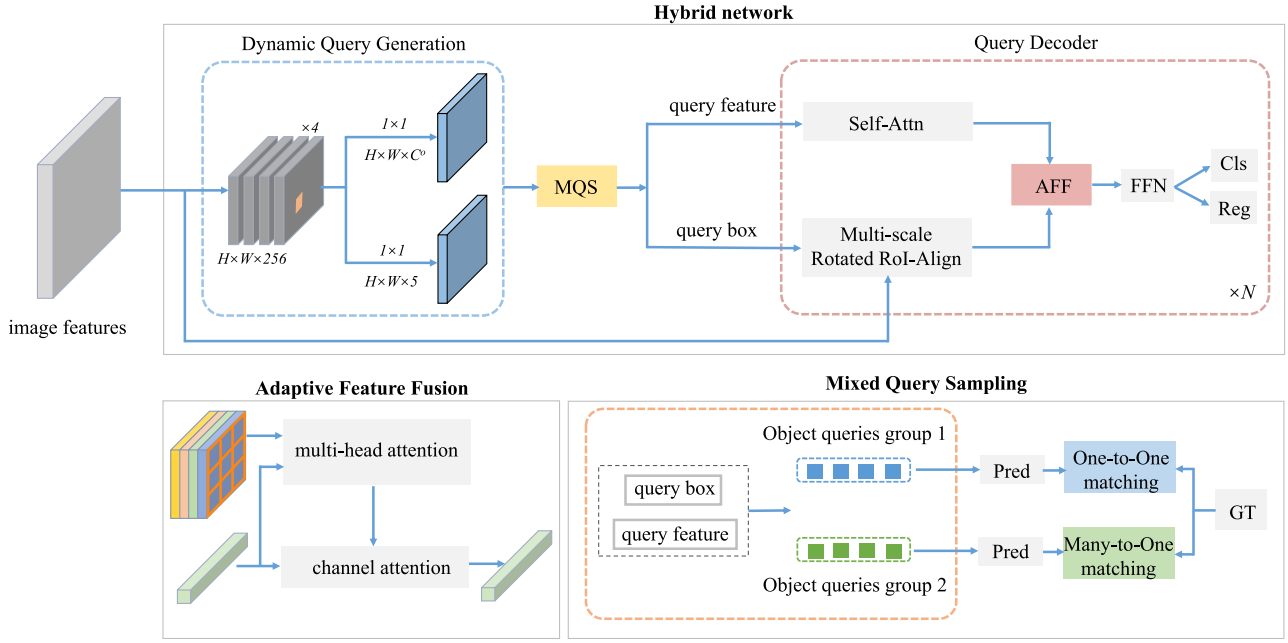


Fig. 3. Architecture of our RHNet. The image features are extracted from the FPN (P_2–P_5). The hybrid network contains a DQG module to produce dynamic queries based on the input image and a query decoder to refine the object queries. The AFF module is designed to effectively aggregate the query feature and multiscale RoI features from the image. During training, an MQS strategy is adopted to select two different groups of object queries (including query boxes and query features) for one-to-one and many-to-one matching, respectively. Self-Attn represents the self-attention module, and FFN denotes the feedforward network. C^o represents the number of object categories.

correct predictions, especially for objects with arbitrary orientations and large aspect ratios. In contrast, sparse detectors apply one-to-one label assignment to a sparse set of object queries, eliminating the need for NMS. However, the learnable queries are difficult to accurately locate aerial objects with highly diverse orientations. Hence, we propose a hybrid network to combine the merits of these two detection paradigms, which is composed of a DQG module and a query decoder.

1) Dynamic Query Generation: The query generation module aims to predict a series of candidate boxes covering the foreground. To reduce the number of parameters, the network uses several shared 3×3 convolutional layers followed by two separate 1×1 convolutions for classification and box regression (see Fig. 3), respectively. The image features adopts P_2–P_5 from FPN, where L represents the pyramid level and P_L indicates the feature map at level L with a stride of 2^L. For each location on P_L, the network predicts the classification scores and box offsets. To decode the oriented boxes, the box offsets are described by 5-D parameters (l, t, r, b, θ) where l, t, r, b denote the distance between the location and each side of the OBB, and $\theta \in (0°, 90°]$ represents the angle of the OBB, as depicted in Fig. 4(b). Each of these four types of distance vectors corresponds to one quadrant of the Cartesian coordinate system. The classification scores represent the probability that the predicted box belongs to each category.

During training, we use an object-adaptation label assignment strategy [23] to define the positive and negative samples, which is more suitable for aerial object with arbitrary shapes and orientations. Specifically, for a sample i located at (x, y), we first define the quality $Q_{i,n}$ for predicting instance n as

$$Q_{i,n} = p^{\gamma}_{i,c^{(n)}} \cdot \left(\mathrm{IoU}\left(\hat{B}_i, B_n\right)\right)^{(1-\gamma)} \cdot G_n(\mathbf{x}). \tag{1}$$

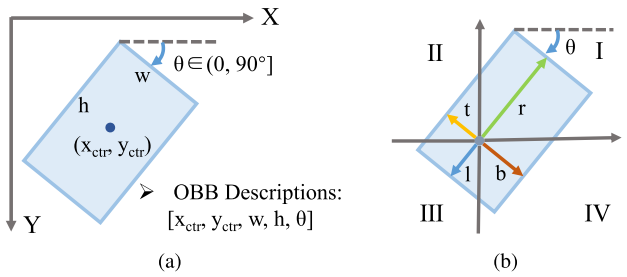


Fig. 4. OBB descriptions. (a) Each OBB is represented by a 5-D vector $(x_{\mathrm{ctr}}, y_{\mathrm{ctr}}, w, h, \theta)$, where $(x_{\mathrm{ctr}}, y_{\mathrm{ctr}})$ refers to the central location and w, h, θ represent the width, width, height, and angle of the OBB, respectively. θ is the acute angle between the OBB and the x-axis ranging in $(0°, 90°]$, and we regard the corresponding side as w of OBB. (b) Illustration of the predicted OBB on locations of the feature map. Specifically, l, t, r, b denote the distance from current location to four sides of the predicted OBB. θ represents the OBB's angle.

$p_{i,c^{(n)}}$ indicates the probability that sample i belongs to the category $c^{(n)}$ of instance n. The intersection over union (IoU) $\mathrm{IoU}(\hat{B}_i, B_n)$ between the predicted OBB $\hat{B}_i$ and ground-truth (GT) OBB B_n reflects the localization accuracy of the predicted OBB. $\gamma \in [0, 1]$ is a hyperparameter used to balance the weights of classification and regression. $\mathbf{x}$ represents the coordinate (x, y), and $G_n(\mathbf{x})$ is a 2-D oriented Gaussian distribution reflecting the prior spatial information of sample i regarding oriented instance n. As shown in Fig. 4(a), the OBB B_n can be denoted as $(c^{(n)}, x^{(n)}_{\mathrm{ctr}}, y^{(n)}_{\mathrm{ctr}}, w^{(n)}, h^{(n)}, \theta^{(n)})$, where $(x^{(n)}_{\mathrm{ctr}}, y^{(n)}_{\mathrm{ctr}})$ refers to the center location, and $w^{(n)}, h^{(n)}, \theta^{(n)}$ represent the weight, height, and angle of the OBB, respectively. In general, for a 2-D Gaussian distribution, the probability density function can be formulated as

$$G_n(\mathbf{x}) = \frac{\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)}{2\pi |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \tag{2}$$

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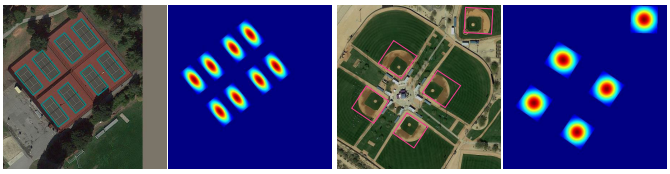


Fig. 5. Illustration of 2-D oriented Gaussian heatmaps for aerial objects with arbitrary shapes and orientations.

where μ represents the mean vector, and Σ denotes the covariance matrix. Thus, each OBB can be converted into a 2-D oriented Gaussian distribution $\mathcal{N}(\mu)$, where

$$\begin{aligned} \mu &= \left(x_{\text{ctr}}^{(n)}, y_{\text{ctr}}^{(n)}\right), \quad \Sigma = \mathbf{R}\Lambda\mathbf{R}^{\top} \\ \mathbf{R} &= \begin{bmatrix} \cos\theta^{(n)} & -\sin\theta^{(n)} \\ \sin\theta^{(n)} & \cos\theta^{(n)} \end{bmatrix} \\ \Lambda &= \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} = \begin{bmatrix} \left(\frac{w^{(n)}}{2}\right)^2 & 0 \\ 0 & \left(\frac{h^{(n)}}{2}\right)^2 \end{bmatrix} \end{aligned} \tag{3}$$

$\mathbf{R}$ denotes the rotation matrix, and Λ denotes the diagonal matrix of eigenvalues λ_1 and λ_2. The density contour of this oriented Gaussian heatmap corresponds to inscribed ellipse of the OBB, as shown in Fig. 5. For locations outside the inscribed ellipse of OBB, their values are set to 0. We see that the values of locations closer to the object's center are higher, and vice versa.

Based on the quality $Q_{i,n}$, we can further derive the soft weight of each sample, which is accommodated with the shape and orientation of aerial object. For more details, refer to [23]. Finally, the loss function for this query generation module can be formulated as

$$\mathcal{L}_{\text{qg}} = \hat{L}_{\text{cls}} + \lambda_{\text{reg}}\hat{L}_{\text{reg}} \tag{4}$$

where $\hat{L}_{\text{cls}}$ is the loss for classification, and $\hat{L}_{\text{reg}}$ is the loss for oriented box regression. λ_{reg} is the balance weight and is set to 1 by default. Quality focal loss (QFL) [33] is adopted for classification, which estimates IoU between the predicted boxes and GT, and thus, the confidence score can better reflect the localization accuracy of predicted boxes. The regression uses rotated IoU loss [34] which is appropriate for oriented object detection.

To obtain a sparse set of object queries, we need to select only a few proposals that are highly relevant to the foreground objects from predictions of the query generation network. These queries are used in the subsequent decoder and have a great impact on final detection performance. As the classification scores cannot perfectly represent the localization accuracy of predicted boxes, the top-K selection method based on the scores is likely to be suboptimal. Moreover, one-to-one matching assigns different labels to the queries, even if they are similar. To avoid unstable network training, the queries are expected to be distinct from each other. Therefore, inspired by the study in [25], we perform NMS on predictions across all the levels from the query generation network, and then keep 300 proposals as the query boxes, if not specified otherwise. To ensure the distinctness of queries, we choose a low IoU threshold. Correspondingly, we gather the query feature for each query box from feature maps of the last 3×3 convolutional layer. These selected queries covering the foreground objects contain rich spatial and content information of the instances, which are beneficial for accurate detection.

It is worth noting that our method differs from traditional detectors that only apply NMS during inference as an additional postprocessing step. Instead, we use NMS in both training and inference to effectively select distinct queries for subsequent one-to-one matching. These selected queries will be further refined through the subsequent query decoder, allowing us to leverage the benefits of NMS while mitigating its negative impact.

2) Query Decoder: The query decoder is adopted to cast the object queries to potential location of objects. As depicted in Fig. 3, each query box is used as a region proposal to extract the corresponding RoI feature from the feature pyramid via the RoI Align operation. Then the query features interact with the corresponding RoI features to output the bounding boxes and category labels accordingly. To achieve better performance, the decoder uses a cascade structure with multiple identical stages, where each stage contains a feature fusion module and a detection head. The feature fusion module refines the proposal features by interacting them with RoI features. Then the detection head with two task-specific layers is applied to produce the final predictions. During training, the decoder adopts set prediction loss to find an optimal one-to-one assignment between the predictions and GT objects. After that, the detection loss is calculated based on the assignment results to optimize the network.

B. Adaptive Feature Fusion

In general, sparse detectors require strong feature representation to cast the object queries to potential location of objects. However, there are many objects with arbitrary orientations and large-scale variations in aerial images, increasing the difficulty of robust feature extraction. To address this issue, we propose an AFF module to better exploit multiscale context information of the rotated regions. Specifically, we first extract multiscale rotational RoI features of the query box from FPN, and then aggregate the information of these features adaptively according to specific query feature.

The detailed structure of AFF is illustrated in Fig. 6. Suppose that there are a set of N object queries, including query boxes $B \in \mathbb{R}^{N\times 5}$ and query features $Q \in \mathbb{R}^{N\times C}$, where C is the feature dimension. Given the feature pyramid $\{P_L\}_{L=2}^{5}$ from FPN, we map each query box $b_n \in \mathbb{R}^{1\times 5}$ to the feature maps at all the levels, and then obtain the rotation-invariant features $f_n^L \in \mathbb{R}^{k\times k\times C}$ from each level by a Rotated RoI Align operation. The feature $f_n \in \mathbb{R}^{k\times k\times 4C}$ is obtained by concatenating RoI features of different scales in channel dimensions. Next, we aggregate the valuable information across different spatial locations based on the query feature $q_n \in \mathbb{R}^{1\times C}$ via a multihead attention mechanism. Here, the query features are obtained from the self-attention module to model the relationships between queries. With H attention heads, this mechanism produces the output feature $o_n \in \mathbb{R}^{1\times 4C}$

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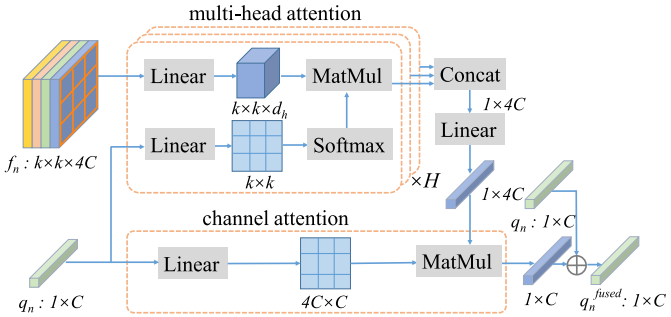


Fig. 6. Detailed architecture of the AFF module. The multiscale RoI feature f_n is first aggregated over different spatial locations via a multihead attention mechanism, and then fused across different scales via a channel attention module. The attention weights are generated based on the query feature q_n.

by computing weighted summation of RoI features f_n over different locations $j = \{1, \ldots, k \times k\}$ inside the RoI region

$$\begin{aligned} \mathbf{o}_n &= \text{Concat}\left(\text{head}_1^n, \ldots, \text{ head }_H^n\right) W^O \\ \text{head}_h^n &= \sum_{j=1}^{k\times k} \text{softmax}(T_h(\mathbf{q}_n))^* \left(\mathbf{f}_n W_h^V\right) \end{aligned} \tag{5}$$

where the projection matrices $W^O \in \mathbb{R}^{4C\times 4C}$ and $W_h^V \in \mathbb{R}^{4C\times d_h}$. T_h is implemented with a fully connected layer followed by a softmax function to produce one spatial weight map $W_h^S \in \mathbb{R}^{k\times k}$. We set the number H of attention heads to 8 in this work and $d_h = 4C/H$. After that, we use a channelwise attention module to adaptively integrate the features of different scales by modeling their interchannel relationships. The fused feature q_n^{fused} of each object query is obtained by

$$\mathbf{q}_n^{\text{fused}} = \mathbf{q}_n + \text{ReLU}(\text{BN}(\mathbf{o}_n * T_c(\mathbf{q}_n)) \tag{6}$$

where T_c is a fully connected layer, and $T_c(q_n) \in \mathbb{R}^{4C\times C}$ reduces channel dimension to the same as the input query feature. Next, the fused features are sent to the feedforward network (FFN) and two task-specific branches for producing final predictions. The classification branch is a fully connected layer to output the probability that the query belongs to K_c classes. And the regression branch is a three-layer perception as in Sparse R-CNN to produce the 5-D offsets with regard to query boxes.

As illustrated in Fig. 7, there exists a spatial misalignment between query box and GT OBB. Our AFF uses a multihead attention mechanism that produces spatial weight maps to aggregate the valuable information across different spatial locations, thus concentrating on the most relevant information for detection. Moreover, considering the large-scale variations in objects in aerial images, such as the soccer ball field (green box) and small vehicles (purple boxes) in Fig. 7(a), our AFF effectively leverages the multiscale rotational RoI features to improve the feature representations, which uses a channelwise attention module to adaptively integrate the features of different scales. Therefore, the AFF can promote the detection performance of our method.

C. Mixed Query Sampling

During training, sparse object detectors adopt one-to-one label assignment that assigns only one positive sample to each

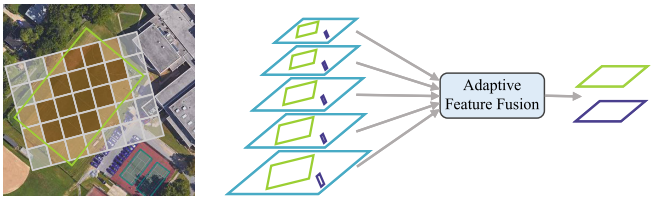


Fig. 7. (a) Illustration of spatial misalignment between query box (white box) and GT OBB (green box). Brown color represents spatial attention map for different locations. And the darker the color, the greater the weight. (b) Illustration of multiscale feature fusion. Green box and purple box denote the soccer ball field and small vehicle, respectively.

GT object, thus eliminating the need for NMS postprocessing. However, it also limits the number of positive samples used in training. Due to the highly diverse orientations of objects in aerial images, it is difficult to produce accurate oriented boxes without sufficient supervision signals for the regression task. Therefore, motivated by the recent studies [28], [29], we attempt to adopt many-to-one assignment to assist network training. Specifically, we find that the expected queries for one-to-one and many-to-one matching are different, and thus design an MQS strategy to combine the advantages of these two assignment schemes.

1) One-to-One Assignment: The object queries obtained from the DQG module are fed to the query decoder for subsequent detection. The decoder contains S cascaded stages with identical structures, and we perform prediction on each stage, respectively. After that, we perform bipartite matching between the predictions and GT objects over each stage. Specifically, suppose there are M GT $G = \{g_1, \ldots, g_M\}$ and N object queries $\widetilde{Q} = \{\widetilde{q}_1, \ldots, \widetilde{q}_N\}$, we can get N detection boxes $D^s = \{d_1^s, \ldots, d_N^s\}$ from each stage s. With one-to-one label assignment, the predictions D^s and GT G are bipartitely matched with a minimum cost. As a result, we can define the optimization target as

$$\begin{aligned} &\mathcal{L}\left(\mathbf{D}^s, \mathbf{G}\right) \\ &= \min_{\pi} \sum_{m=1}^{M+1} \sum_{n=1}^{N} C(m,n) * \pi(m,n) \\ &\text{s.t.} \sum_{m=1}^{M+1} \pi(m,n) = 1, \quad \sum_{n=1}^{N} \pi(m,n) = s_m, \quad \sum_{m=1}^{M+1} s_m = N \\ &\qquad \pi(m,n) \geq 0, \quad m = 1, 2, \ldots, M+1, \quad n = 1, 2, \ldots, N \end{aligned} \tag{7}$$

where m, n represent the indices of GT and detection boxes, respectively. Note that $M + 1$ here represents the background class, and $\pi(m, n)$ represents the matching result between each GT m and detection box n. s_m denotes the number of detection boxes assigned to each GT m. For one-to-one matching, s_m can be written as

$$s_m = \begin{cases} 1, & m \leq M \\ N - M, & m = M + 1 \end{cases} \tag{8}$$

and the rest $N - M$ boxes are regarded as negative samples. The cost is $C(m, n)$ if GT m is matched with detection box n,

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which is defined as the weighted summation of classification and regression loss

$$C(m,n)=\begin{cases}\mathcal{L}_{\text{cls}}(m,n)+\alpha*\mathcal{L}_{\text{reg}}(m,n), & m\le M\\ \mathcal{L}_{\text{cls}}(\text{background},n), & m=M+1\end{cases} \tag{9}$$

where $\mathcal{L}_{\text{cls}}$ denotes the focal loss [15] for classification, and $\mathcal{L}_{\text{reg}}$ denotes rotated IoU loss for oriented box regression. The hyperparameter α is used to balance the loss terms and is set to 2.5 empirically. As for negative samples that only involve classification, α equals to 0. To solve this bipartite matching problem, we use the Hungarian algorithm following DETR [6]. After getting the assignment results, the training loss can be calculated by

$$\mathcal{L}_{\text{one2one}}=\sum_{s=1}^{S}\mathcal{L}_{\text{Hungarian}}\left(\mathbf{D}^s,\mathbf{G}\right). \tag{10}$$

2) Many-to-One Assignment: Using multiple positive samples for each GT is beneficial to the detection task, especially for oriented box regression. In this work, we attempt to apply many-to-one assignment to another group of queries and use auxiliary training loss to supervise network training. Specifically, we adopt an advanced optimal transport assignment (OTA) method [35] for many-to-one matching. In this way, the bipartite matching is extended to an optimal transport problem between the predictions and GT. Given another group of queries $\hat{Q}=\{\hat{q}_1,\ldots,\hat{q}_N\}$, we can obtain N detection boxes $\hat{D}^s=\{\hat{d}_1^s,\ldots,\hat{d}_N^s\}$ from each stage s. The difference between OTA and bipartite matching is the number of positive samples s_m assigned to each GT m. For many-to-one assignment, s_m can be written as

$$s_m=\begin{cases}k_m, & m\le M\\ N-\sum_{m=1}^{M}k_m, & m=M+1.\end{cases} \tag{11}$$

Following OTA, we use dynamic estimation to determine the number k_m based on the IoU between predictions and GT. Formally, we sum up top T IoU value of N detection boxes for each GT m, and then convert it into an integer

$$k_m=\left\lfloor\sum\text{top}_T\{\text{IoU}(m,n)|\ n=1,2,\ldots,N\}\right\rfloor. \tag{12}$$

Here, we use $T=10$ by default. In practice, we adopt the Sinkhorn–Knopp algorithm to solve this optimal transport problem. After obtaining the matching results, the auxiliary training loss can be formulated by

$$\mathcal{L}_{\text{many2one}}=\sum_{s=1}^{S}\mathcal{L}_{\text{Sinkhorn-Knopp}}\left(\hat{D}^s,\mathbf{G}\right). \tag{13}$$

3) Mixed Query Sampling: As stated in Section III-A, proper object queries are essential for subsequent label matching and detection. To ensure a high recall, object queries need to cover as many objects in the image as possible. As one-to-one matching assigns only one positive sample to each GT, the matched queries have different labels. To avoid unstable training, these queries are expected to be distinct from each other, as shown in Fig. 8(a). We perform NMS on predictions of the DQG module to select the expected queries and use a low NMS threshold to make sure the queries are distinct. Nevertheless, for many-to-one matching, we will assign multiple positive samples to each GT, and hence, the queries assigned to the same GT should be similar to some degree, as described in Fig. 8(b). Therefore, we adopt a high NMS threshold to reserve some duplicate outputs of the same target. It is worth noting that these two groups of object queries are fed to the same decoder for further refinement, and the differences between them lie in the label assignment strategy and optimization targets. This MQS strategy can generate two different sets of queries for one-to-one and many-to-one assignments, respectively, allowing the detector to leverage the merits of these two schemes simultaneously. We conduct experiments in ablation study to verify the effectiveness of this strategy.



(a) (b)

Fig. 8. Illustration of different strategies for label assignment. (a) Example of one-to-one matching. There is only one positive sample assigned to each GT object, and hence, the queries are expected to cover the foreground objects and be distinct from each other. (b) Example of many-to-one matching. Multiple positive samples are assigned to each GT object, and thus, the queries assigned to the same target are similar to some degree. Red boxes are GTs, and other boxes are query boxes.

The total training loss function consists of three parts as follows:

$$\mathcal{L}_{\text{total}}=\lambda_q\mathcal{L}_{qg}+\lambda_o\mathcal{L}_{\text{one2one}}+\lambda_m\mathcal{L}_{\text{many2one}} \tag{14}$$

where λ_q, λ_o, and λ_m are the scaling factors. We set $\lambda_q=1$, $\lambda_o=1$, and $\lambda_m=0.5$ in this work. During inference, we use a low NMS threshold to select the object queries, which is consistent with query sampling for one-to-one matching during training. The final detections are obtained from the query decoder without postprocessing. Note that NMS cannot guarantee to perfectly filter out false-positive predictions while retaining all the true positives. Therefore, we only use NMS to select high-quality object queries, which will be further refined in subsequent stages.

IV. Experiments

A. Experimental Setting

1) Datasets: The proposed RHNet is evaluated on several public aerial object detection datasets, including DOTA1.0 [10], HRSC2016 [11], and UCAS-AOD [12]. DOTA1.0 is widely used for detecting multiclass oriented objects. In total, there are 2806 images in this dataset ranging from 800 × 800 to 4000 × 4000 pixels in size, which are

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TABLE II
ABLATION STUDIES ON DIFFERENT COMPONENTS ON THE DOTA1.0 VALIDATION SET

Methods	Dynamic Query Generation	Adaptive Feature Fusion	Mixed Query Sampling	mAP(%)	FPS
FCOSR (w/ NMS)	✓	✗	✗	68.12	**14.1**
Sparse R-CNN (S=6)	✗	✗	✗	52.99	13.6
	✗	✓	✗	61.98	9.5
	✗	✗	✓	60.86	13.6
	✗	✓	✓	65.35	9.5
RHNet (S=2)	✓	✗	✗	71.26	13.2
	✓	✓	✗	71.95	11.7
	✓	✗	✓	72.18	13.2
	✓	✓	✓	**72.85**	11.7

collected from multiple sensors and platforms. And there are 188 282 objects of 15 categories annotated with OBBs. The dataset contains 1411 images used for training, 458 images for validation, and 937 images used for testing without any annotations. The ablation study is performed on this validation set. With a stride of 824, the original images are cropped into 1024×1024 patches. When compared with other methods, the training and validation sets are both used for training, while the rest is used for testing. And the stride of the crop operation is set to 512. The HRSC2016 dataset is used for ship detection with 436 training, 181 validation, and 444 test images. And the size of the images varies between 300×300 and 1500×900. Also, we train the models with the training and validation sets and test with the rest images. UCAS-AOD contains 1510 aerial images that are used to detect small objects such as cars and airplanes. With the ratio of 5:2:3, the images are randomly divided into three groups, i.e., training set, validation set, and test set.

Unless otherwise specified, we use mean average precision (mAP) under the IoU threshold of 0.5 as the evaluation metric, which is defined the same as PASCAL VOC 2007 [36] object detection challenge.

2) Implementation Details: In our experiments, ResNet-50 [37] pretrained on ImageNet [38] is used as the backbone network if not specified. We use AdamW [39] optimizer with a weight decay of 0.0001 in all the experiments, and the mini-batch size is 8 on four GPUs (two per GPU). The learning rate starts from 2.5×10^{-5} and is multiplied by 0.1 at each decay step. For both training and inference, we resize the images to 1024×1024. The number of object queries sampled from the query generation network is set to 300. Random horizontal flipping and random rotation are used to augment the training data. The total epochs of models trained on the DOTA1.0 dataset are set to 36, and we adjust the learning rate at 24 and 33 epochs. Besides, the models on the HRSC2016 and UCAS-AOD datasets are optimized for 5.5k steps with learning rates changing at 3.7 and 5k steps. With a batch size of 1, the inference is performed on a single 2080Ti GPU.

B. Ablation Study

To verify the effectiveness of our method, we conduct a series of experiments on the DOTA1.0 validation set.

1) Ablation Studies on Importance of Each Component: We compare the proposed RHNet with FCOSR [5] and Sparse R-CNN [7] in terms of different components. During training, FCOSR adopts the object-adaptation label assignment strategy [23] to handle arbitrary-oriented objects. And we also adapt Sparse R-CNN to aerial images for detecting oriented objects. As shown in Table II, FCOSR with NMS achieves 68.12% mAP. In practice, the structure of detection head in FCOSR is the same as the DQG module, which uses dense prior knowledge to locate the objects with large variations. Besides, Sparse R-CNN with six cascaded stages achieves an mAP of 52.99%, lower than FCOSR. This is mainly because the learnable queries cannot accurately locate the arbitrary-oriented objects.

The proposed RHNet uses a hybrid network to combine the advantages of dense and sparse detectors. Specifically, as it contains a DQG module to provide better initial object queries, the performance of the detector is greatly improved from 52.99% to 71.26% mAP, compared with Sparse R-CNN. Meanwhile, our RHNet outperforms FCOSR with NMS, due to the refinement of the query decoder. This is because the query decoder adopts one-to-one matching to eliminate the negative impact caused by NMS. Note that we only use two cascaded stages in our experiments considering training efficiency.

After that, we integrate the AFF module and MQS strategy into our hybrid network. We see that AFF can improve the performance by 0.69% mAP, demonstrating that strong feature representation is beneficial for promoting the detection performance. Besides, AFF leads to a significant improvement over Sparse R-CNN. This is because the learnable queries in Sparse R-CNN are more inaccurate and the proposed AFF can generate more effective features that concentrate on the most relevant information for detection. In addition, with the help of MQS, the performance can be further promoted by 0.92% mAP. This is because increasing the number of positive samples provides more supervisory signals for oriented box regression. It is worth mentioning that the proposed AFF and MQS greatly improve the performance of Sparse R-CNN. However, due to the lack of dense prior information, it still achieves inferior performance compared with the proposed RHNet.

2) Effectiveness of DQG: We further analyze the effectiveness of the designed DQG and compare the results of different label assignment strategies. First, we compare the performance of FCOSR equipped with the original fixed

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TABLE III
ABLATION EXPERIMENTS WITH DQG

Methods	Fixed label assignment	QGM	DQG	mAP(%)	Training time
FCOSR (w/ NMS)	✓	✗	✗	66.91	8.6 min
	✗	✓	✗	65.85	40.2 min
	✗	✗	✓	68.12	10.2 min
RHNet (S=2)	✓	✗	✗	67.73	9.7 min
	✗	✓	✗	68.14	42.4 min
	✗	✗	✓	71.26	11.1 min

TABLE IV
ABLATION EXPERIMENTS WITH MQS

Query sampling methods		Recall	Label assignment	mAP(%)
top-K	$K = 300$	72.24	one-to-one	59.26
	$K = 1800$	94.80		68.74
NMS	$\tau = 0.1$	87.61		71.30
	$\tau = 0.2$	88.34		70.15
	$\tau = 0.3$	88.92		70.52
	$\tau = 0.4$	89.80		69.87
	$\tau = 0.5$	90.32		69.35
top-K	$K = 300$	72.24	many-to-one	62.59
	$K = 1800$	94.80		65.82
NMS	$\tau = 0.1$	87.61		69.75
	$\tau = 0.2$	88.34		70.34
	$\tau = 0.3$	88.92		70.86
	$\tau = 0.4$	89.80		71.45
	$\tau = 0.5$	90.32		71.62

label assignment strategy [16], QGM in [25], and our DQG, respectively. Next, we apply them to our hybrid network. As shown in Table III, our DQG achieves better performance than QGM and FCOSR with fixed label assignment. This is because we use a more appropriate object-adaptation label assignment strategy to accommodate arbitrary shapes and orientations of aerial objects. Moreover, from Table III, we see that QGM takes much longer training time (almost four times) than our DQG. This is mainly caused by the bipartite matching (implemented on CPU) directly on dense feature maps. Besides, the proposed hybrid network consistently improves detection performance compared with FCOSR, demonstrating the effectiveness of our method.

3) Effectiveness of MQS: To validate the effectiveness of the MQS strategy, we conduct an ablation analysis with different query sampling methods, and the results are given in Table IV. As we can see, query sampling with NMS obtains a higher recall than the top-K method with the same number of queries, thus leading to better final detection performance. Although increasing the number of object queries in top-K improves the recall, it also brings a lot of computational costs and reduces the inference speed. Meanwhile, we compare the performance of different label assignment schemes under various NMS thresholds τ ranging from 0.1 to 0.5. As presented in Table IV, for one-to-one matching, $\tau = 0.1$ works better, while $\tau = 0.5$ works best for many-to-one matching. The difference between optimal τ values validates that these two label assignment schemes need to use different NMS thresholds for query sampling, which leads to more stable training of the detector. We observe that decreasing τ will benefit one-to-one assignment, suggesting that distinct queries are more helpful to the network training as each GT corresponds to only one positive sample. On the other hand, increasing τ is good for many-to-one assignment, indicating that queries assigned to the same GT should be similar. As a result, we choose $\tau = 0.1$ for one-to-one assignment and $\tau = 0.5$ for many-to-one assignment in our experiments. Fig. 9 visualizes some example images of the generated proposals for these two label assignment schemes and final detection results. For one-to-one matching, the proposals of each object are distinct from each other. For many-to-one matching, each object is surrounded by multiple sample proposals.

TABLE V
ABLATION EXPERIMENTS WITH NUMBER OF STAGES

	mAP(%)	FPS
$S = 1$	70.68	12.7
$S = 2$	72.85	11.7
$S = 3$	73.16	9.9
$S = 4$	73.49	8.6
$S = 5$	72.54	7.9
$S = 6$	71.28	6.6

TABLE VI
ABLATION EXPERIMENTS WITH NUMBER OF QUERIES

	mAP(%)	FPS
$N = 150$	71.54	12.1
$N = 300$	72.85	11.7
$N = 450$	73.45	10.3
$N = 600$	73.12	9.4

4) Hyperparameter Analysis: We conduct experiments to examine the impact of the number of cascaded stages S and object queries N to model's performance.

a) Number of cascaded stages: According to the results presented in Table V, the detection performance can further increase with more stages, but five or more stages will not bring additional improvements and even lead to a slight decrease. Besides, adding more stages will increase the computational cost, and thus, we choose $S = 2$ to achieve a better performance while maintaining a reasonable inference speed.

b) Number of object queries: The number of object queries may affect the recall rate. As shown in Table VI, as N increases, detection accuracy improves slightly, but inference speed decreases. And there is a slight performance drop when N continues to increase. This is probably because the objects in images are well-covered by a sufficient number of queries. Here, we choose $N = 300$ as it provides a better balance between detection accuracy and speed.

5) Complexity Analysis: In Table VII, we further analyze the detection performance and complexity of the proposed method, including the model parameters, FLOPs, inference speed, and mAP. We add the proposed components gradually and use only two cascaded stages. As depicted in Table VII, the designed DQG of our RHNet brings more computational cost in terms of GFLOPs compared with Sparse R-CNN, which is mainly caused by the convolutional layers. And the

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Fig. 9. Examples of generated proposals and detection results. The first and second rows denote the generated query box with our MQS strategy for one-to-one and many-to-one matching (pink and green boxes), respectively. And the third row is the final detection results for different categories (denoted with different colors). Except for some queries that fall in the background, the queries assigned to foreground objects are distinct for one-to-one matching. While for many-to-one matching, the objects are surrounded by multiple samples. The combination of these two assignment schemes ultimately leads to promising detection results.

TABLE VII

COMPLEXITY ANALYSIS. THE MODELS ARE TRAINED WITH RESNET-50 BACKBONE NETWORK

Methods	DQG	AFF	MQS	Params	GFLOPs	FPS	mAP(%)
FCOSR (w/ NMS)	✓	✗	✗	36.85M	228.39	14.1	68.12
Sparse R-CNN (S=6)	✗	✗	✗	106.54M	126.08	13.6	52.99
RHNet (S=2)	✓	✗	✗	63.29M	236.17	13.2	71.26
	✓	✓	✗	116.97M	252.25	11.7	71.95
	✓	✓	✓	116.97M	252.25	11.7	72.85

TABLE VIII

EVALUATION OF THE PROPOSED MODULES ON HORIZONTAL OBJECT DETECTION TASK OF THE DOTA1.0 VALIDATION SET

Methods	DQG	MQS	mAP(%)
FCOSR (w/ NMS)	✓	✗	68.03
Sparse R-CNN (S=6)	✗	✗	46.91
RHNet (S=2)	✓	✗	69.33
	✓	✓	69.98

query decoder of our RHNet introduces more parameters than FCOSR. As FLOPs is not the only factor that affects the inference speed of a model and the convolution operation has been considerably optimized, our RHNet outperforms FCOSR and Sparse R-CNN while the inference speed drops slightly. In addition, using the MQS strategy facilitates the network training and brings no extra cost during inference.

6) Evaluation on Horizontal Detection: This work primarily focuses on oriented object detection in aerial images. Compared with horizontal objects, oriented objects with arbitrary orientations increase the difficulty of accurate detection. As horizontal object can be regarded as a special case of oriented object when the angle is set to zero, our method may also be applicable in horizontal object detection. The DQG and MQS modules can be adapted to horizontal object detection, so we conduct additional experiments on the horizontal bounding box version of the DOTA1.0 dataset to evaluate their performance. Note that we keep the dataset preprocessing and parameter settings consistent with our other experiments on oriented object detection. The models are trained with the training set and evaluated on the validation set. The results are shown in Table VIII, where the mAP is

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TABLE IX

COMPARISON RESULTS WITH RELATED METHODS ON THE DOTA1.0 TEST SET. EACH COLUMN'S BEST AND SECOND-BEST RESULTS ARE INDICATED BY RED AND BLUE, RESPECTIVELY. RR REPRESENTS RANDOM ROTATION

Methods	Backbone	RR	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP(%)
Two-stage Methods																		
FR-O [10]	ResNet-101		79.09	69.12	17.17	63.49	34.20	37.16	36.20	89.19	69.60	58.96	49.40	52.52	46.69	44.80	46.30	52.93
R-DFPN [40]	ResNet-101		80.92	65.82	33.77	58.94	55.77	50.94	54.78	90.33	66.34	68.66	48.73	51.76	55.10	51.32	35.88	57.94
R^2CNN [41]	ResNet-101	✓	80.94	65.67	35.34	67.44	59.92	50.91	55.81	90.67	66.92	72.39	55.06	52.23	55.14	53.35	48.22	60.67
RRPN [17]	ResNet-101		88.52	71.20	31.66	59.30	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58	61.01
ICN [42]	ResNet-101		81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23	68.16
CSL [43]	ResNet-50	✓	89.09	79.22	45.78	67.94	66.58	56.29	71.55	90.80	80.50	76.61	58.53	61.01	59.92	69.51	57.59	68.73
RoI-Transformer [1]	ResNet-101	✓	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
CAD-Net [2]	ResNet-101		87.80	82.40	49.40	73.50	71.10	63.50	76.70	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
SCRDet [44]	ResNet-50	✓	89.48	79.83	50.43	62.51	76.72	74.31	85.42	90.86	84.19	82.51	42.84	60.15	74.18	70.62	51.81	71.72
Gliding vertex [4]	ResNet-101	✓	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
Oriented R-CNN [18]	ResNet-50		89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
ReDet [19]	ReResNet-50		88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
One-stage Methods																		
PIoU [45]	ResNet-101	✓	80.90	69.70	24.10	60.20	38.30	64.40	64.80	90.90	77.20	70.40	46.50	37.10	57.10	61.90	64.00	60.50
CFC-Net [46]	ResNet-101	✓	89.08	80.41	52.41	70.02	70.02	78.11	87.21	90.89	84.47	85.64	60.51	61.52	67.82	68.02	50.09	73.50
R^3Det [3]	ResNet-101	✓	88.76	83.09	50.91	67.27	76.23	80.39	86.72	90.78	84.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79
S^2A-Net [20]	ResNet-50		89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
R^3Det-GWD [47]	ResNet-50	✓	88.82	82.94	55.63	72.75	78.52	83.10	87.46	90.21	86.36	85.44	64.70	61.41	73.46	76.94	57.38	76.34
R^3Det-KLD [48]	ResNet-50	✓	88.90	84.17	55.80	69.35	78.72	84.08	87.00	89.75	84.23	85.73	64.74	61.80	76.72	78.49	70.89	77.36
Anchor-free Methods																		
IENet [49]	ResNet-101	✓	80.20	64.54	39.82	32.07	49.71	65.01	52.58	81.45	44.66	78.51	46.54	56.73	64.40	64.24	36.75	57.14
Axis Learning [50]	ResNet-101		79.53	77.15	38.59	61.15	67.53	70.49	76.30	89.66	79.07	83.53	47.27	61.01	56.28	66.06	36.05	65.98
DRN [51]	ResNet-101	✓	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
O^2-DNet [52]	ResNet-101		89.31	82.14	47.33	61.21	71.32	74.03	78.62	90.76	82.23	81.36	60.93	60.17	58.21	66.98	61.03	71.04
BBAVectors [53]	ResNet-101		88.35	79.96	50.69	62.18	78.43	78.98	87.94	90.85	83.58	84.35	54.13	60.24	65.22	64.28	55.70	72.32
FCOSR [5]	ResNet-50	✓	89.26	78.18	50.27	70.16	80.63	79.42	88.28	90.84	84.81	86.38	61.11	68.24	74.85	75.36	60.61	75.89
End-to-End Methods																		
Sparse R-CNN [7]	ResNet-50	✓	71.69	54.67	31.63	57.84	67.67	70.77	69.42	74.13	56.70	55.04	31.10	35.86	42.43	37.14	45.11	53.41
O^2DETR [30]	ResNet-50		88.76	81.91	51.20	72.18	77.64	80.47	87.84	90.85	84.56	81.68	61.42	64.61	67.50	64.28	62.15	74.47
DDQ [25]	ResNet-50	✓	88.76	76.72	51.29	69.21	79.69	84.22	88.28	90.82	82.74	84.18	54.67	63.66	77.90	82.80	68.97	76.26
AO2-DETR [31]	ResNet-50		89.27	84.97	56.67	74.89	78.87	82.73	87.35	90.50	84.68	85.41	61.97	69.96	74.68	72.39	71.62	77.73
Ours	ResNet-50		87.76	80.59	51.69	71.06	78.99	80.77	88.27	90.86	80.67	86.22	61.64	63.78	77.34	73.61	56.55	75.49
Ours	ResNet-50	✓	89.22	80.01	53.06	72.99	79.74	80.33	88.47	90.83	82.78	85.22	64.35	64.34	77.61	79.76	73.35	77.47

TABLE X

COMPARISON RESULTS WITH RELATED METHODS ON THE HRSC2016 DATASET

Methods	R^2CNN [41]	RC1 & RC2 [54]	RRPN [17]	YOLOv3-R [55]	R^2PN [56]	RetinaNet-H [3]	RRD [57]	RoI-Transformer [1]	Gliding Vertex [4]	RetinaNet-R [3]
mAP(%)	73.07	75.70	79.08	79.53	79.60	82.89	84.30	86.20	88.20	89.20
Methods	IENet [49]	Axis Learning [50]	CenterNet-R [58]	CSL [43]	BBAVectors [53]	R^3Det [3]	CFC-Net [46]	S^2A-Net [20]	AO2-DETR [31]	Ours
mAP(%)	75.01	78.15	83.72	86.20	88.60	89.30	89.50	90.17	88.12	**90.27**

TABLE XI

COMPARISON RESULTS WITH RELATED METHODS ON THE UCAS-AOD DATASET

Methods	car	airplane	mAP(%)
YOLOv3-R [55]	74.63	89.52	82.08
RetinaNet-R [15]	84.64	90.51	87.57
FR-O [10]	86.87	89.86	88.36
RoI-Transformer [1]	88.02	90.02	89.02
CFC-Net [46]	89.29	88.69	89.49
DAL [21]	89.25	90.49	89.87
Ours	**89.62**	**90.54**	**90.08**

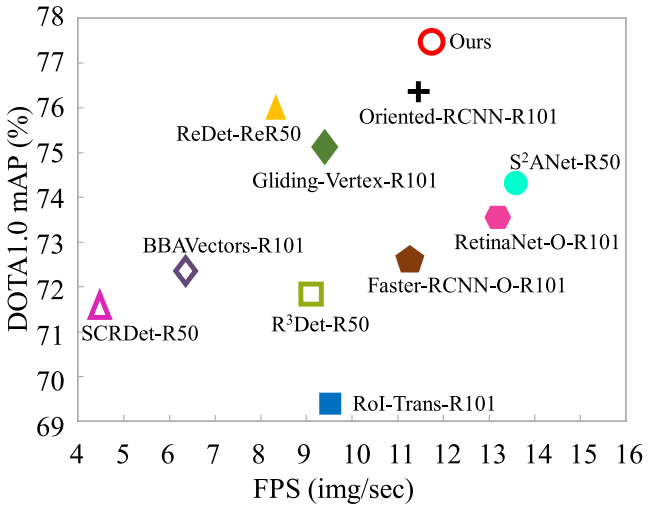


Fig. 10. Accuracy and speed of different methods on the DOTA1.0 test set.

calculated under the IoU threshold of 0.5. We see that our RHNet outperforms FCOS with NMS, thanks to the refinement of the query decoder. Moreover, compared with Sparse R-CNN, our RHNet contains a DQG module that provides better initial object queries, resulting in an improved performance from 46.91% to 69.33% mAP. This indicates that directly applying sparse detectors to aerial images leads to inferior performance. This may be because aerial images contain some objects that are densely distributed, posing a challenge for sparse detectors to learn accurate initial object queries. In addition, the use of MQS further enhances the performance by an additional 0.65% mAP, as it provides more positive training samples to supplement one-to-one matching. Notably,

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Fig. 11. Visualized detection results of our RHNet on the DOTA1.0 test set.

these two modules also bring benefits to horizontal object detection, showcasing the versatility and effectiveness of our method.

C. Comparison With the State-of-the-Art Methods

To further investigate the performance of the proposed RHNet, we compare our method with other state-of-the-art detectors on several public datasets for aerial images, including DOTA1.0, HRSC2016, and UCAS-AOD.

1) Results on DOTA1.0: There are multiple categories of arbitrary-oriented objects in DOTA1.0, making it a challenging dataset to analyze. We divide the comparison methods into four categories, i.e., two-stage, one-stage, anchor-free, and end-to-end methods. We present the results on the DOTA1.0 test set in Table IX. As can be seen, our method attains performance on par with previous works that use only random flipping. Moreover, random rotation further boosts the performance by generating more samples with varying orientations. Thus, our RHNet outperforms all other traditional methods and achieves comparable performance to recent end-to-end method AO2-DETR with fewer training epochs. For example, our RHNet performs better than the two-stage detector Oriented R-CNN (77.47% mAP versus 76.28% mAP). Also, our method outperforms the one-stage detector R^3Det-KLD [48] and anchor-free detector FCOSR [5]. In addition, a comparison of detection accuracy and speed of different methods under identical conditions is shown in Fig. 10. The results demonstrate the superiority of our method to achieve the state-of-the-art performance in aerial images. Besides, we use only single-scale images during training and testing in this work. To further validate the effectiveness of our method, the detection results of some example images are visualized in Fig. 11. We can see that our RHNet is capable of accurately detecting objects with different scales and arbitrary orientations in various scenarios. Moreover, our method can also perform well for the categories with large aspect ratios, such as harbors in the first row. The promising results demonstrate the robustness of our RHNet that can successfully handle all these challenging cases.

2) Results on HRSC2016 and UCAS-AOD: We present the experimental results of some representative methods and our RHNet on the HRSC2016 dataset in Table X. We observe that our RHNet reaches an mAP of 90.27% without using NMS postprocessing, outperforming all other methods. In Fig. 12, we present the detection results of our RHNet on some example images from the HRSC2016 dataset. Despite these ships

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Fig. 12. Visualized detection results of our RHNet on the HRSC2016 dataset.



Fig. 13. Visualized detection results of our RHNet on the UCAS-AOD dataset.



(a) (b) (c)

Fig. 14. Some examples of failure cases. The top row represents the GT, and the bottom row represents the detection results of our method. Boxes with different colors denote objects of different categories. (a) Objects with extreme aspect ratios. (b) Small objects. (c) False positives.

having different sizes, large aspect ratios, and arbitrary orientations, our method still exhibits outstanding performance, owing to the specially designed components for oriented objects at various scales. Besides, Table XI shows the comparison results on the UCAS-AOD dataset. Compared with other methods, RHNet also consistently yields the best performance of 90.08% mAP. Fig. 13 visualizes some detection results on UCAS-AOD.

D. Analysis of Failure Cases

We show some examples of failure cases in Fig. 14. Although our method achieves superior performance, there are still some false detection results, such as the inaccurate detections of objects with extreme aspect ratios in Fig. 14(a) and false negatives of small objects in Fig. 14(b). Besides, the number of object queries is fixed regardless of the input image and is usually larger than the number of GTs, which inevitably incurs some false positives, as shown in Fig. 14(c). These cases are still very challenging for object detection in aerial images, and we will further research on these issues in our future work.

V. Conclusion

In this article, we develop an effective RHNet for detecting rotated objects at various scales in aerial images. Specifically, we use a DQG module to generate a few high-quality oriented queries. These queries contain sufficient positional and content information of foreground oriented objects. Next, the queries are fed to subsequent query decoder for further refinement, which removes NMS postprocessing via a one-to-one matching loss. Moreover, an AFF module is proposed to learn stronger feature representations of multiscale rotated regions. In addition, we design an MQS strategy that integrates many-to-one label assignment into the training process of the detector. This enables the regression branch to benefit from using more training samples, which can produce more accurate oriented boxes. Experiments on three benchmark aerial datasets verify the superior performance of our method.

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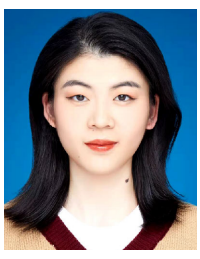
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